Name: _____ Date: _____

Coordinate Geometry & Transformations: Color By Number

Color Code:

_____ Gray (−5, 2)	_____ Red (−4, 1)	_____ Brown (4, 5)	_____ Yellow (1, −2)	_____ Light Blue (6, 2)
_____ Black (4, 1)	_____ Purple (2, 3)	_____ Orange (3, 2)	_____ Green (−3, 2)	_____ Tan (−4, 10)

Name: _____ Date: _____

Coordinate Geometry & Transformations: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Translate the point (2, 3) using the rule (x, y) → (x + 4, y − 1). Find the image.

1. Color: _____

2.
Reflect the point (3, −2) across the x-axis. Find the image.

2. Color: _____

3.
Reflect the point (−4, 5) across the y-axis. Find the image.

3. Color: _____

4.
Rotate the point (2, 3) 90° counterclockwise about the origin. Find the image.

4. Color: _____

5.
Rotate the point (4, −1) 180° about the origin. Find the image.

5. Color: _____

6.
Dilate the point (3, −6) by a scale factor of $\frac{1}{3}$ with the center at the origin. Find the image.

6. Color: _____

7.
Dilate the point (−2, 5) by a scale factor of 2 with the center at the origin. Find the image.

7. Color: _____

8.
Reflect the point (1, 4) across the line y = x. Find the image.

8. Color: _____

9.
The translation (x, y) → (x − 3, y + 2) moves point A to A′. If A = (5, 1), find A′.

9. Color: _____

10.
Rotate the point (−2, −5) 90° clockwise about the origin. Find the image.

10. Color: _____